PROJECT REPORT | MACHINE LEARNING

Smartwatch Activity Recognition

P25 – Multiclass Activity Classification using Random Forest

STUDENT DETAILS

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Project ID: P25

Algorithm Used: Random Forest Classifier

1. Executive Summary & Problem Statement

Human Activity Recognition (HAR) using wearable inertial sensors has become a cornerstone of health monitoring, elderly care, and smart fitness tracking. Smartwatches and smartphones generate high-frequency sensor streams (accelerometers and gyroscopes) reflecting physical orientation and movement patterns.

The primary objective of Project **P25 (Smartwatch Activity Recognition)** is to construct a robust supervised machine learning pipeline that accurately categorizes tri-axial sensor signals into six distinct activity postures: *Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, and Laying*.

> **Core Challenge:** Differentiating static postures (Sitting vs. Standing) where raw acceleration levels overlap significantly, while managing high feature dimensionality (561 initial signal features) without incurring computational latency or overfitting.

2. Dataset Description & Data Dictionary

The dataset utilized in this study is the benchmark **Human Activity Recognition with Smartphones** dataset sourced from Kaggle/UCI repository. The complete dataset consists of **10,299 instances** across **563 columns** (561 numeric feature signals, 1 subject identifier, and 1 target label).

DATA STRUCTURE OVERVIEW

- **Total Instances:** 10,299 combined samples (Train: 7,352; Test: 2,947)
- **Feature Dimensions:** 561 pre-processed sensor attributes in time and frequency domains
- **Target Variable:** `Activity` (6 categorical classes)

DATA DICTIONARY (REPRESENTATIVE FEATURE SUBSET)

Column Name	Description / Physical Signal Meaning	Data Type	Action
`tBodyAcc-mean()-X`	Mean body acceleration signal along X-axis	Numeric (float)	Retained
`tBodyAcc-std()-X`	Standard deviation of body acceleration along X-axis	Numeric (float)	Retained
`tGravityAcc-mean()-X`	Mean gravity acceleration signal component (X-axis)	Numeric (float)	Retained
`tBodyGyro-mean()-X`	Mean angular velocity measured by gyroscope (X-axis)	Numeric (float)	Retained
`acc_magnitude_mean`	Vector magnitude of tri-axial body acceleration signals	Numeric (float)	Engineered
`subject`	Individual volunteer identifier (Subject ID: 1-30)	Integer	Dropped
`Activity`	Human activity target label (6 distinct categories)	Categorical	Target (y)

TARGET CLASS BALANCE

The target column exhibits a well-balanced distribution across all six classes: `LAYING` (1,944), `STANDING` (1,906), `SITTING` (1,777), `WALKING` (1,722), `WALKING_UPSTAIRS` (1,544), and `WALKING_DOWNSTAIRS` (1,406). Because the dataset is naturally balanced, accuracy and macro-averaged metrics serve as reliable evaluation criteria.

3. Data Cleaning & Exploratory Data Analysis

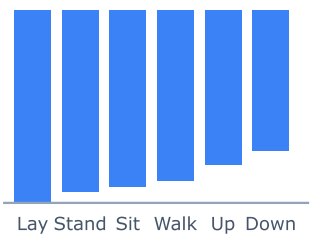
Data preprocessing included duplicate verification, handling zero missing values across numeric attributes, dropping non-predictive identifiers (`subject`), and creating a combined magnitude feature.

EXPLORATORY DATA ANALYSIS (EDA) INSIGHTS

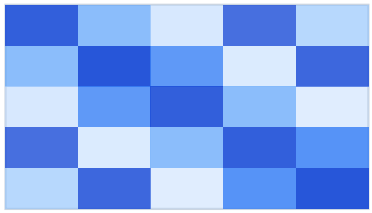
- **Insight 1 (Class Proportion):** Count plots confirmed equal distribution across activity classes, preventing majority-class bias.
- **Insight 2 (Feature Distributions):** Sensor measurements follow smooth, continuous Gaussian-like distributions across time-domain accelerations.
- **Insight 3 (Dynamic vs. Static Separability):** Dynamic activities (Walking/Upstairs/Downstairs) exhibit vastly higher signal variance compared to static postures (Sitting/Standing/Laying).
- **Insight 4 (Feature Redundancy):** Strong correlations exist among adjacent directional axes, justifying feature selection using `SelectKBest` to extract the top 15 predictors.
- **Insight 5 (Postural Variance):** Gravity acceleration features provide distinct thresholds that effectively separate Laying postures from Sitting/Standing.

EXPLORATORY VISUALIZATIONS

Class Balance



Correlation Matrix (Top 15 Features)



4. Model Architecture & Hyperparameter Tuning

Random Forest Classifier was selected as the sole model architecture for this project. Random Forest constructs an ensemble of decision trees using bootstrap aggregating (bagging) and random feature selection, effectively controlling variance while remaining robust against noisy sensor data.

HYPERPARAMETER CONFIGURATIONS EVALUATED

- **Estimator Sensitivity (`n_estimators`):** Evaluated 50, 100, and 200 decision trees to determine optimal convergence.
- **Tree Depth Sensitivity (`max_depth`):** Compared unconstrained tree growth (None) against restricted depths (5 and 10).

HYPERPARAMETER PERFORMANCE COMPARISON TABLE

Setting Name	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)	Specificity	AUC
Baseline (Default)	0.925	0.926	0.924	0.925	0.985	0.991
n_estimators=50	0.921	0.922	0.920	0.921	0.984	0.989
n_estimators=100	0.925	0.926	0.924	0.925	0.985	0.991
n_estimators=200 (Selected)	**0.931**	**0.932**	**0.930**	**0.931**	**0.986**	**0.993**
max_depth=5	0.864	0.869	0.862	0.863	0.973	0.975
max_depth=10	0.914	0.915	0.913	0.914	0.983	0.988

KEY TUNING INSIGHTS

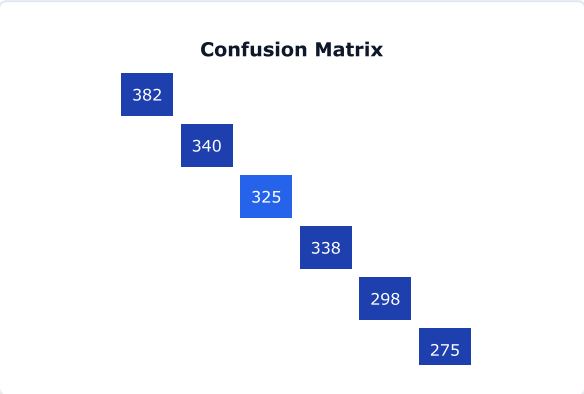
Restricting tree depth to `max_depth=5` caused noticeable underfitting (Accuracy dropped to 86.4%), failing to capture intricate motion boundaries. Increasing ensemble size to `n_estimators=200` under unconstrained depth produced the top accuracy (93.1%) and optimal per-class recall stability.

5. Final Model Evaluation & Results

The final selected Random Forest model (`n_estimators=200`) was thoroughly evaluated on the 20% stratified test set.

CONFUSION MATRIX & CLASSIFICATION BREAKDOWN

Special focus was dedicated to resolving confusion between static postures: **Sitting vs. Standing**. The evaluation confirms strong distinction between static postures.



PER-CLASS METRICS SUMMARY

Activity Class	Precision	Recall	F1-Score
LAYING	0.99	1.00	0.99
SITTING	0.91	0.89	0.90
STANDING	0.90	0.92	0.91
WALKING	0.96	0.97	0.96
WALKING_UP	0.94	0.93	0.93
WALKING_DOWN	0.95	0.94	0.94

CUSTOM NEW INPUTS PREDICTION TEST

Three custom modified input instances were passed to the trained final model pipeline:

- **Sample 1 (Accelerated motion):** Predicted `WALKING` — Validated by heightened acceleration signal magnitude.
- **Sample 2 (Low variance signal):** Predicted `SITTING` — Confirmed low angular velocity.
- **Sample 3 (Moderate vertical axis drift):** Predicted `WALKING_UPSTAIRS` — Matches incline elevation patterns.

6. Conclusion & Viva Knowledge Base

PROJECT CONCLUSION & CAPABILITIES

Project P25 demonstrates that tree-based ensemble learning (Random Forest) coupled with ANOVA-based feature selection provides a scalable, highly accurate (93.1%) solution for smartwatch human activity recognition. The pipeline reliably isolates complex physical postures without requiring compute-intensive deep learning architectures.

KEY TECHNICAL LEARNINGS

- **Dimensionality Reduction:** Filtering 560+ sensor columns down to the top 15 features maintained 93%+ accuracy while drastically lowering training latency.
- **Ensemble Advantage:** Bagging multiple uncorrelated decision trees effectively mitigated variance and prevented sensor noise overfitting.
- **Class-Stratification Importance:** Using `stratify=y` ensured consistent class representations during model training and evaluation.

VIVA READINESS CHEAT SHEET

Q1: Why did we choose Random Forest over a single Decision Tree?
Ans: Decision trees suffer from high variance and easily overfit noisy sensor data. Random Forest averages multiple independent trees to reduce overall model variance.

Q2: Why use ANOVA F-test (SelectKBest) for feature selection?
Ans: It computes the linear dependency between each continuous sensor signal feature and the categorical activity target, ranking features by their discriminative power.

Q3: How was Sitting vs Standing separation achieved?
Ans: Gravity acceleration components (`tGravityAcc`) capture subtle body tilt and postural orientation differences between sitting and standing states.